Internal Pressure of a Carbonated Beverage Can

Kashyap, Anuj
Pittala, Ithishree
Howell, Liliana

Section 519L 04/15/2026

Abstract—**This experiment estimated the internal pressure of an unopened carbonated beverage can using two independent methods. The first method used two strain gauges aligned with the principal directions of the can wall to measure hoop and longitudinal strain before and after opening. These strain changes were converted to stress using plane-stress relationships and then related to pressure using thin-walled pressure-vessel theory. The second method estimated pressure from the change in can diameter after opening, treating the diameter contraction as circumferential strain. The strain-gauge method yielded a longitudinal stress of around 43.6 MPa and a hoop stress of about 110 MPa, which corresponded to pressure estimations of 270 kPa and 340 kPa, respectively. A strain-gauge pressure estimate of 317 kPa was obtained by combining the two stress components. An average pressure estimate of 305.6 ± 55.9 kPa was obtained using the diameter-change approach. Although electrical noise limited the strain-gauge method and the diameter method's small diameter change in relation to caliper resolution limited the diameter method, both approaches yielded comparable pressure estimates, supporting the application of thin-walled pressure-vessel theory for estimating soda-can pressure.**

Index Terms—**Internal pressure, Strain gauge, Thin-walled pressure vessel, Quarter Wheatstone bridge Circuit**

I. Introduction

THIS experiment estimated the internal pressure in an unopened carbonated beverage can using strain measurements and thin-walled pressure vessel theory. A second pressure estimate was also required without using strain gauges so that the pressure estimates could be compared for consistency and sensitivity to measurement uncertainty. Since the wall thickness of a soda can is much smaller than its diameter, the can may be modeled as a thin-walled cylindrical pressure vessel. Under this assumption, stress through the wall thickness is neglected and the wall is treated as a plane-stress element. Internal pressure in a cylindrical vessel produces two principal stresses in the wall: hoop stress, σ_H, which acts circumferentially, and longitudinal stress, σ_L, which acts parallel to the axis of the cylinder. These stresses describe how the can carries pressure and provide the basis for relating the measured strain to the internal pressure [1], [2]. Two separate strain gauges were installed in the middle of the can's surface on opposing sides for this experiment. The longitudinal strain was measured by aligning the 0° gauge with the can's vertical axis, and the hoop strain was recorded by aligning the 90° gauge with the circumferential direction. Instead of requiring a whole strain-transformation approach, the observed strains could be employed immediately in the pressure-vessel analysis since these directions match the primary directions of an ideal thin-walled cylinder. This configuration allowed one or two gauges to be purposefully aligned with principal strains in accordance with the project guidelines [1], [2].

For a thin-walled cylindrical pressure vessel, the hoop and longitudinal stresses are related to internal pressure, P (kPa), mean can diameter, d (mm), and wall thickness, t (mm), by (1) and (2).

$$\sigma_H = \frac{Pd}{2t} \tag{1}$$

$$\sigma_L = \frac{Pd}{4t} \tag{2}$$

These relationships demonstrate that for a perfect thin-walled cylinder, the hoop stress is double the longitudinal stress (Fig. 1). Both stresses can be merged into a single pressure expression because they are both caused by the same internal pressure. When the thin-wall pressure relation is substituted with the major stresses, it yields (3).

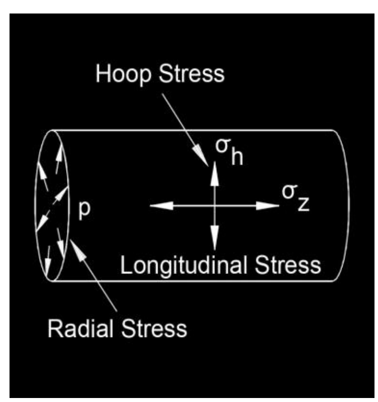

Fig 1. Hoop, longitudinal, and radial stress directions in a cylindrical pressure vessel subjected to internal pressure [4].

$$P = \frac{(\sigma_H + \sigma_L)4t}{3d} \quad (3)$$

Equation (3) is useful because it allows the internal pressure to be found once the two in-plane stresses in the can wall are known [1], [2].

Strain was not directly measured by the strain gauges as a physical displacement. Instead, as the can wall deformed, each strain gauge acted as a variable resistor whose resistance changed with the wall strain. Because this resistance change was small, each gauge was connected through a quarter Wheatstone bridge and amplifier circuit before being read by the DAQ. The amplified bridge voltage output was then converted to strain using the LabVIEW VI. The measured strain was calculated from the amplified bridge voltage, V_{amp} (V), excitation voltage, V_s (V), amplifier gain, G, nominally 900, and gauge factor, G_f, equal to 4. For each saved LabVIEW data point, a sample frame of 2500 voltage samples were used, and the VI recorded the mean and standard deviation of that frame for uncertainty analysis. For the final project, ten stored data points were collected both before and after the can was opened so that the strain change was based on averaged voltage states rather than individual instantaneous readings. Fig. 2 shows the quarter Wheatstone bridge, amplifier, and DAQ signal path used for the strain-gauge measurements.

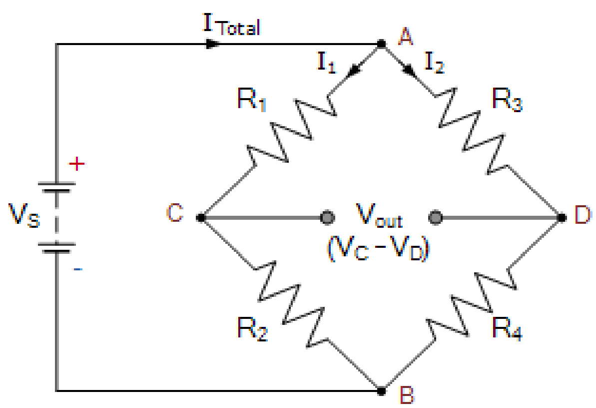

Fig 1. Excitation V_s, bridge output ΔV_g, amplifier output V_{amp}, and DAQ connections [6].

$$\varepsilon = \frac{4\Delta V_{amp}}{G V_s G_f} \quad (4)$$

The strain was measured both before and after the can was opened to air pressure because the soda can experiment is a one-shot measurement. Consequently, the strain change employed in the analysis was (5).

$$\Delta\varepsilon = \varepsilon_f - \varepsilon_o \quad (5)$$

Where ε_f is the final strain following opening and ε_o is the original strain in the sealed can. The longitudinal strain (ε_L) and hoop strain (ε_H) required for the pressure-vessel analysis were calculated from the measured strain variations for the current gauge setup.

To convert the measured strains into stresses, the can wall was treated as an isotropic material under plane stress. With Young's modulus, E (Pa), and Poisson's ratio, ν, the hoop and longitudinal stresses are given by (6) and (7).

$$\sigma_H = \frac{E}{1-\nu^2} \cdot (\varepsilon_H + \nu\varepsilon_L) \quad (6)$$

$$\sigma_L = \frac{E}{1-\nu^2} \cdot (\varepsilon_L + \nu\varepsilon_H) \quad (7)$$

These relations allow the directly measured strains to be converted into the principal stresses required in (3). In this way, the strain recorded by the two orthogonally aligned gauges could be used to estimate the original internal pressure of the unopened can. A second non-strain-gauge method was also developed so that both pressure estimates could be compared in the final analysis [1], [2].

The change in can diameter before and after opening was used in the second pressure estimate. The can wall contracted somewhat and the interior pressure fell to air pressure when the can was opened. The diameter change was used as an approximation of the hoop strain because this contraction happened in a circumferential direction. If D_f is the final open-can diameter and D_o is the initial closed-can diameter, the diameter-based circumferential strain ε_D is (8).

$$\varepsilon_D = \frac{D_o - D_f}{D_o} \quad (8)$$

This strain can be used in conjunction with the thin-walled pressure-vessel stress relations and the plane-stress form of Hooke's law to estimate pressure because it corresponds to hoop strain. The diameter method's pressure is (9).

$$P_D = \frac{2Et\varepsilon_D}{D_o(1-\frac{\nu}{2})} \quad (9)$$

Where t is the thickness of the can wall, ν is Poisson's ratio, E is Young's modulus, and P_D is the diameter-method pressure estimate. An independent pressure estimate that could be compared with the strain-gauge method was produced by this method.

The DAQ records V_{amp} as an analog voltage, which is then converted into digital values using an analog-to-digital converter (ADC). ADC bit depth and the selected input range (sample window). The SADI DAQ used in this course has 14-bit ±10 V ADC inputs with configurable gain and sample rate. [7] A 14-bit ADC, commonly referred to as the least significant bit (LSB), is used to quantify the selected input window. This is the voltage step's size. [6]

$$Resolution = \frac{V_{max} - V_{min}}{2^n - 1} \quad (10)$$

Where V_{max} and V_{min} are the sample-window limits (V) and n is the ADC bit depth (bits). The method yields sample windows of ±0.64 V, ±1.28 V, ±2.56 V, ±5.12 V, and ±10.24 V. A saturation check was used to determine the smallest window that remained unsaturated under the maximum anticipated loads. [6]

II. Procedure

For this experiment, a Coca-Cola soda can was prepared with two strain gauges that were adhered to the surface of the can and allowed to cure for 24 hours. Gauge wires were then soldered onto the cured strain gauges and tested using a multimeter to ensure proper electrical connections. A Data Acquisition Device (DAQ) and Wheatstone Bridge (WSB) were then utilized to log the voltages obtained from the soda can.

The main specimen prepared for this study was the soda can and all of its associated attachments. After obtaining a soda can, a caliper was used to record its diameter and length, while a micrometer was used to measure its thickness. A digital scale was also used to determine the mass of the can. After collecting all physical measurements, the strain gauges were applied. Two strain gauges were mounted at the center of the can surface on opposite sides, with the 0° gauge aligned with the vertical axis to measure longitudinal strain and the 90° gauge aligned with the circumferential direction to measure hoop strain. The surface area for the strain gauges was then prepared for application. Using low-grit sandpaper, the locations where the strain gauges were to be attached were sanded and cleaned (Fig. 3).

Fig. 3. The image shows the sanding of the Soda Can and the tape aligning the strain gauge

Using the previously measured length from the calipers, two pieces of tape were placed latitudinally and longitudinally at half the length of the can. To ensure that the tape was not applied at an angle, the printed lines and features on the can were used as reference points. These pieces of tape were then used to ensure that the strain gauges were perpendicular to each other. This method ensured perpendicular alignment of the strain gauges, which allows for the measurement of principal stresses.

The strain gauges were then handled using clear tape and aligned with the latitudinal and longitudinal reference tape. Once properly aligned, the tape was slowly lifted to allow the super glue to be applied beneath each strain gauge. After applying the adhesive and allowing it to bond and partially cure, the tape was removed and the adhesive was allowed to fully cure. This process was repeated for the second strain gauge. Once the strain gauges were mechanically secured, the gauge wires were soldered onto the strain gauges and connected to the DAQ. A multimeter was then used to verify proper electrical continuity (Fig. 4).

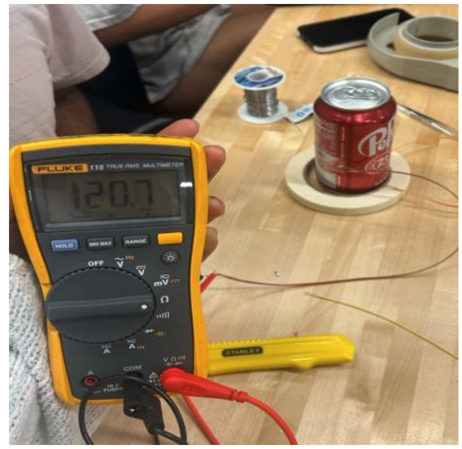

Fig. 4. Multimeter used to ensure proper connections

Before data collection, the Wheatstone Bridge (WSB) and Virtual Instrument (VI) were constructed. The Wheatstone Bridge was first connected to the amplifier by attaching two wires from the WSB. The WSB output was then connected to the IN+ and IN− terminals of the amplifier PCB so that the bridge voltage, V_g, could be accurately measured. The positive and negative B terminals were then connected to the DAQ to measure the excitation voltage, V_s. The amplifier was powered using a 3.3 V supply and ground, and all data was read through the DAQ (Fig. 5). LabVIEW VI then utilized the WSB and DAQ configuration.

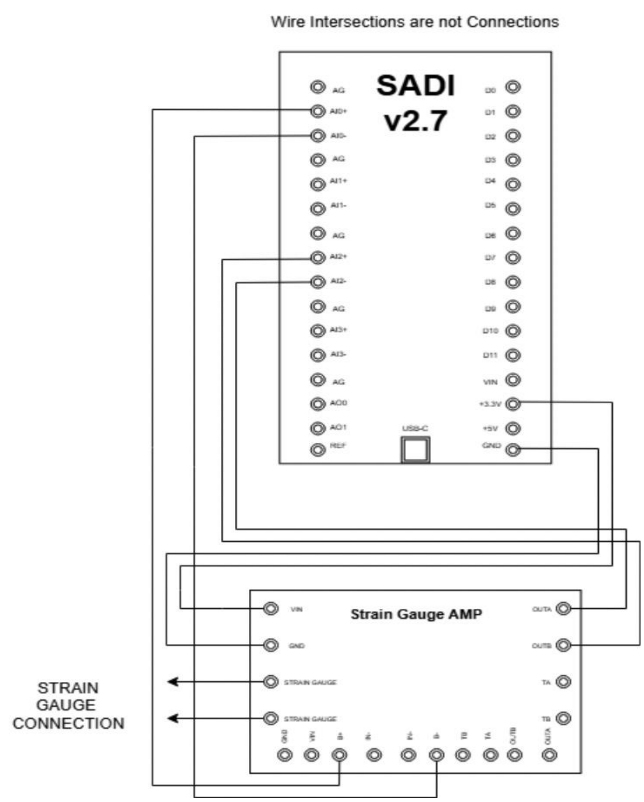

Fig. 5. An image displaying the DAQ wiring and logic [6]

A main VI and a sub-VI were then developed to calculate the principal stresses. The main VI provided strain output from the measured voltage signals, and those strain values were then used with the relevant material properties and can dimensions to calculate stress and pressure Then the outputs of the main VI are used to calculate the principal stresses. This is accomplished by first determining the longitudinal and latitudinal stresses, as shown in (6) and (7), and then calculating the pressure using (3) After connecting both DAQs to the VI, data collection was initiated. Prior to recording results, testing was conducted to ensure that appropriate sampling windows were selected for V_s, V_g, and V_{amp} to prevent signal saturation. The windows selected were $\pm$ 2.56 V for the V_s, V_g, and V_{amp} channels. Once channels were selected, both VIs were executed, and 10 data points were recorded to measure strain on the can. The can was then shaken and the diameter was recorded. After all the proper measurements have been taken, the can is opened, and 10 data points were recorded to measure the strain after opening. The standard deviation and mean were then calculated for use in uncertainty analysis to find the 95% confidence interval (CI), where $\bar{x}$ represents the sample mean, t is the t-value from Student's t-distribution, s is the standard deviation, and N is the sample size (11).

$$CI = \bar{x} \pm t \frac{s}{\sqrt{N}} \tag{11}$$

III. Results

There were two methods used to calculate pressure. The first method utilized strain gauges, while the second involved measurements of the can's diameter. The body of the soda can was composed of Aluminum 3104-H19, which allowed for the determination of essential material properties such as the Modulus of Elasticity (69 GPa) and Poisson's Ratio (0.34) [3]. All relevant physical dimensions were then measured using calipers for the length and diameter, and a micrometer was used to measure the thickness after sectioning the soda can in half, these measurements were then recorded (Table I).

TABLE I
BASELINE MEASUREMENTS FOR SODA CAN

Symbol	Quantity	Value
T	Can Thickness	0.1016 mm
D	Can Diameter	65.77 mm
l	Can Length	12.59 cm
E	Elastic Modulus	69 GPa
v	Poisson's Ratio	0.34

For the first method, two strain gauges were applied to measure strain along the longitudinal and circumferential axes. Using the VI, V_{amp}, standard deviation (std. dev.), and corresponding strain values were recorded ten times for the unopened can. Using the calculated strain values and previously recorded physical properties, the stress along the horizonal and vertical axes were calculated using strain from both axes and recorded in Tables II (6) and III (7).

TABLE II
UNOPENED VERTICAL STRAIN GAUGE DATA

Iteration	V_{amp} [V]	Std. dev. V_{amp} [V]	o	Stress [kPa]
1	-0.59 ±0.00748	0.0038	-0.000510 ±0.0000443	-40200.00 ± 619.00
2	-0.59 ±0.00769	0.0039	-0.000510 ±0.0000468	-40200.00 ± 619.00
3	-0.593 ±0.00750	0.0038	-0.000510 ±0.000045	-40200.00 ± 619.00
4	-0.593 ±0.00766	0.0039	-0.000510 ±0.0000459	-40200.00 ± 619.00
5	-0.593 ±0.00761	0.0039	-0.000510 ±0.0000471	-40100.00 ± 616.00
6	-0.593 ±0.00758	0.0039	-0.000510 ±0.0000459	-40100.00 ± 616.00
7	-0.593 ±0.00744	0.0038	-0.000510 ±0.0000468	-40100.00 ± 616.00
8	-0.592 ±0.00778	0.0040	-0.000510 ±0.000474	-40100.00 ± 616.00
9	-0.593 ±0.00737	0.0038	-0.000510 ±0.0000459	-40100.00 ± 616.00
10	-0.593 ±0.00739	0.0038	-0.000510 ±0.0000462	-40100.00 ± 616.00

TABLE III
UNOPENED HORIZONTAL STRAIN GAUGE DATA

Iteration	V_{amp} [V]	Std. dev. V_{amp} [V]	Strain [mm/mm]	Stress [kPa]
1	0.015 ±0.00756	0.00386	0.000014 ±0.00122	-12440.00 ± 1420.00
2	0.0163 ±0.00769	0.00392	0.000014 ±0.00124	-12440.00 ± 1420.00
3	0.016 ±0.00761	0.00388	0.000014 ±0.00123	-12440.00 ± 1420.00
4	0.0169 ±0.00772	0.00394	0.000014 ±0.00125	-12440.00 ± 1420.00
5	0.0172 ± 0.00762	0.00389	0.000015 ±0.00123	-12360.00 ± 1410.00
6	0.0179 ±0.00768	0.00392	0.000015 ±0.00124	-12360.00 ± 1410.00
7	0.0178 ±0.00760	0.00388	0.000015 ±0.00123	-12360.00 ± 1410.00
8	0.0172 ±0.00758	0.00387	0.000015 ±0.00125	-12360.00 ± 1410.00
9	0.0177 ±0.00775	0.00396	0.000015 ±0.00125	-12360.00 ± 1410.00
10	0.0181 ±0.00764	0.00390	0.000015 ±0.00123	-12360.00 ± 1410.00

After recording and analyzing the 10 data points from the soda can's strain gauges and calculating the stress experienced by the can along both axes, the can is opened. After opening the can, additional data points are recorded to measure the voltage difference and strain. This allows for the calculation of the stress experienced by the can along both axes with results in Tables IV and V.

TABLE IV
OPENED VERTICAL STRAIN GAUGE DATA

Iteration	V_{amp} [V]	Std. dev. V_{amp} [V]	Strain [mm/mm]	Stress [kPa]
1	-0.4934 ±0.0075	0.0038	-0.00042 ±0.00005	5428.88 ± 619.00
2	-0.4931 ±0.00754	0.0038	-0.00042 ±0.00005	5428.88 ± 619.00
3	-0.4932 ±0.00762	0.0039	-0.00042 ±0.00005	5428.88 ± 619.00

TABLE IV CONTINUED
OPENED VERTICAL STRAIN GAUGE DATA

Iteration	V_{amp} [V]	Std. dev. V_{amp} [V]	Strain [mm/mm]	Stress [kPa]
4	-0.4933 ±0.00757	0.0039	-0.00042 ±0.00005	5428.88 ± 619.00
5	-0.4921 ±0.00742	0.0038	-0.00042 ±0.00005	5428.88 ± 619.00
6	-0.4933 ±0.00732	0.0037	-0.00042 ±0.00005	5428.88 ± 619.00
7	-0.4928 ± 0.00741	0.0038	-0.00042 ±0.00005	5428.88 ± 619.00
8	-0.4919 ±0.00765	0.0039	-0.00042 ±0.00005	5401.04 ± 616.00
9	-0.4920 ± 0.00749	0.0038	-0.00042 ±0.00005	5401.04 ± 616.00
10	-0.4928 ± 0.00741	0.0038	-0.00042 ±0.00005	5401.04 ± 616.00

TABLE V
OPENED HORIZONTAL STRAIN GAUGE DATA

Iteration	V_{amp} [V]	Std. dev. V_{amp} [V]	Strain [mm/mm]	Stress [kPa]
1	1.6353 ± 0.00762	0.0039	0.00140 ±0.00001	97695.39 ± 11137.00
2	1.6351 ±0.00789	0.0040	0.00140 ±0.000012	97695.39 ± 11137.00
3	1.6349 ±0.00766	0.0039	0.00140 ±0.000013	97695.39 ± 11137.00
4	1.6347 ±0.00769	0.0039	0.00140 ±0.00001	97695.39 ± 11137.00
5	1.6347 ±0.00781	0.0040	0.00140 ±0.000011	97695.39 ± 11137.00
6	1.6348 ±0.00777	0.0040	0.00140 ±0.000012	97695.39 ± 11137.00
7	1.6355 ±0.00757	0.0039	0.00140 ±0.000011	97695.39 ± 11137.00
8	1.6343 ±0.00781	0.0040	0.00139 ±0.00001	97617.37 ± 11128.00
9	1.6343 ±0.00774	0.0040	0.00139 ±0.000012	97617.37 ± 11128.00
10	1.6341 ±0.00753	0.0038	0.00139 ±0.000011	97617.37 ± 11128.00

After calculating the stress of the can on both axes, the pressure of the can before and after it was opened can be calculated (3) and results are displayed in Table VI.

TABLE VI
HORIZONTAL STRAIN GAUGE PRESSURE

Iteration	Pressure (unopened) [kPa]	Pressure (opened) [kPa]
1	-108.47 ± 1.55	212.60 ± 11.16
2	-108.47 ± 1.55	212.60 ± 11.16
3	-108.47 ± 1.55	212.60 ± 11.16
4	-108.47 ± 1.55	212.60 ± 11.16
5	-108.25 ± 1.54	212.60 ± 11.16
6	-108.25 ± 1.54	212.60 ± 11.16
7	-108.25 ± 1.54	212.60 ± 11.16
8	-108.25 ± 1.54	212.38 ± 11.15
9	-108.25 ± 1.54	212.38 ± 11.15
10	-108.25 ± 1.54	212.38 ± 11.15

The pressure of the can before and after opening was then used to determine the change in internal pressure and results are displayed in Table VII. This was accomplished by subtracting the pressure measured after the can was opened from the pressure measured prior to opening.

TABLE VII
CHANGE IN VAMP AND PRESSURE

Iteration	$\Delta V_{amp, horizontal}$ [V]	$\Delta V_{amp, vertical}$ [V]	ΔPressure [kPa]
1	1.6194±0.01063	0.0996±0.0107	-321.07±11.275
2	1.6188 +0.0077	0.0992 ±0.00755	-321.07±11.274
3	1.6181 ±0.0075	0.0998±0.00763	-321.07±11.273
4	1.6178±0.00766	0.0999±0.00757	-321.07±11.274
5	1.6176±0.00761	0.1010±0.00742	-320.85±11.275
6	1.6169±0.00758	0.1000±0.00733	-320.85±11.274
7	1.6177±0.00758	0.1003±0.00742	-320.85±11.274
8	1.6171±0.00778	0.1006±0.00766	-320.63±11.275
9	1.6166±0.00737	0.1011±0.0075	-320.63±11.274
10	1.6160±0.00026	0.0999 ±0.00742	-320.63±11.274

The resulting difference represents the initial internal pressure of the sealed can, where an average of 320.87 ± 11.27 kPa was found.

Table VIII displays the results of stress and pressure calculated from average hoop and longitudinal strain calculated from the difference of unopened and opened strains (6) and (7).

TABLE VIII
RESULTS FROM AVERAGE HOOP AND LONGITUDINAL STRAIN

Hoop/Longitudinal	Stress [MPa]	Pressure [kPa]
Hoop	110.05 ± 18.34	340.02 ± 104.68
Longitudinal	43.62 ± 9.32	270.00 ± 78.61

In the second method, the change in pressure was calculated using the change in diameter of the can. To ensure an adequate sample size, the diameter was recorded ten times under three different conditions: when the can was initially received, after the can was shaken, and after the can was opened. After recording these measurements, the diameters before and after opening, along with v and E were used to calculate the internal pressure (9). The resulting pressure values were then recorded and summarized in Table IX.

TABLE IX
CHANGE IN DIAMETERS

Iteration	Unopened Diameter [mm]	Shaken Can Diameter [mm]	Opened Diameter [mm]	Pressure [kPa]
1	65.0 ± 0.005	65.0 ± 0.005	65.0 ± 0.005	186.8 ± 65.4
2	63.0 ± 0.005	65.0 ± 0.005	63.0 ± 0.005	282.3 ± 98.8
3	65.0 ± 0.005	65.0 ± 0.005	64.0 ± 0.005	359.55 ± 125.8
4	65.0 ± 0.005	65.0 ± 0.005	64.0 ± 0.005	283.1 ± 99.1

TABLE IX CONTINUED
CHANGE IN DIAMETERS

Iteration	Unopened Diameter [mm]	Shaken Can Diameter [mm]	Opened Diameter [mm]	Pressure [kPa]
5	65.0 ± 0.005	65.0 ± 0.005	64.0 ± 0.005	243.4 ± 85.2
6	65.2 ± 0.005	65.3 ± 0.005	64.0 ± 0.005	485.301 ± 169.9
7	65.1 ± 0.005	65.9 ± 0.005	64.5 ± 0.005	243.396 ± 85.2
8	64.9 ± 0.005	60.0 ± 0.0056	64.0 ± 0.005	367.348 ± 128.6
9	65.4 ± 0.005	64.9 ± 0.005	64.5 ± 0.005	361.753 ± 126.6
10	65.1 ± 0.005	65.5 ± 0.005	64.5 ± 0.005	243.396 ± 85.20
AVG.	65.0 ± 0.0005	65.2 ± 0.0005	64.3 ±0.005	305.63 ± 55.94

The average strain from the diameter measurements found using (8) was 0.0116 ± 0.0109 mm/mm and used to calculate the pressure using (9).

To compare the data measured from the experiment, a class data set of 24 was documented and analyzed. The analysis was conducted for both the full data set and a reduced data set of 21 after removing outliers (Table X).

TABLE X
OVERALL CLASS DATA

Sample Size	Average Pressure [kPa]	Std. dev. [kPa]	95% CI [kPa]
24	255.211	125.782	53.113
21*	290.709	87.221	39.702

Additionally, the internal pressure can also be determined by using the individual strain measurements obtained from both the hoop and axial strain gauges. By analyzing these separate strain components, an alternative calculation method can be applied to estimate the pressure more comprehensively and verify the accuracy of the results. Table XI displays the values from hoop strain and Table XII displays the values from axial strain. Both were calculated after removing outliers.

TABLE XI
HOOP CLASS DATA

Sample Size	Average Pressure [kPa]	Std. dev. [kPa]	95% CI [kPa]
13	288.335	98.082	59.705

TABLE XII
AXIAL CLASS DATA

Sample Size	Average Pressure [kPa]	Std. dev. [kPa]	95% CI [kPa]
8	294.566	70.525	58.960

IV. Discussion

Strain Gauge Method

The objective of this experiment was to determine the internal pressure of a carbonated beverage can using strain measurements and thin-walled pressure vessel theory. A stain gauge method was used where two strain gauges were positioned 90 degrees from one another, one in the hoop and the other in the longitudinal direction. This was done because they correspond to the principal strain directions in a thin-walled cylindrical pressure vessel. Since the principal directions were already known, a strain rosette was not needed. A rosette is typically used when principal directions are unknown and implemented incorrectly can lead to error. So, in this case, using two gauges in the principal directions reduces error and simplifies calculations. Using this method, the calculated hoop stress was approximately 110.05 ± 18.34 MPa and the longitudinal stress was approximately 43.62 ± 9.32 MPa. These values corresponded to the internal pressure estimates of 340.02 kPa ± 104.68 from hoop stress and 270.00 ± 78.61 kPa from longitudinal stress. Using (3), a resultant pressure of 317 kPa was found. The difference between these pressure values reflects experimental uncertainty as well as limitations in the theoretical assumptions.

The hoop-based pressure is considered more reliable because the measured hoop strain was significantly higher than the longitudinal strain, resulting in a higher signal to noise ratio. The longitudinal strain was very small in magnitude, making it highly sensitive to electrical noise and amplification error which affected the resolution of the voltage signal. Therefore, small variations in the measured voltage produced relatively large percentages errors in the longitudinal strain and the corresponding pressure calculation.

The strain gauge method included uncertainty from the digital data acquisition devices. Strain was calculated from amplified voltage signals using a quarter Wheatstone bridge and amplifier. The relatively large standard deviation in the amplified voltage indicated that electrical noise contributed to measurement variability. This noise propagates through the strain calculations and affects the final stress and pressure values largely.

The measured strain did not decrease after opening because the measured voltage increased after opening. Although the physical strain in the can wall decreases when internal pressure is released, the measured strain may increase due to the sign convention and calibration of the instrumentation. For this reason, the analysis relied on the change in strain magnitude between the unopened and opened states rather than the absolute direction of the measurement.

Shaking the can temporarily disturbs the carbonation equilibrium and therefore increases the diameter, but after waiting sufficient time, the internal pressure returns approximately to its equilibrium value. This is due to the re-establishment of gas-liquid equilibrium between dissolved CO_2 and the headspace, assuming the temperature remains constant.

Diameter Change Method

A second method was used to estimate pressure by measuring the change in can diameter before and after opening. In this approach the strain was approximated using (8) and this strain was substituted into the thin-walled pressure vessel relations to solve for pressure (9). This method does not require strain gauges or electrical instrumentation and instead relies on direct geometric measurements. It provided an alternative experimental approach and serves as a comparison to the strain gauge method. This method resulted in an average internal pressure of 305.63 ± 55.94 kPa.

This method is limited by measurement resolution. The expected change in diameter due to internal pressure is extremely small, often on the order of micrometers, which approaches or falls below the resolution limit of standard calipers. As a result, the uncertainty in the diameter measurements was large relative to the actual deformation, leading to variability in the calculated pressure results.

Comparison of Methods

While both methods rely on the assumption that the can behaves as a thin-walled cylindrical pressure vessel, the aluminum beverage can have non-uniform geometry, including curved end caps and thickness variations due to manufacturing processes. These features result in non-uniform stress distributions that are not captured by the idealized equations. Measurements were taken in the middle of the can to minimize these effects. Additionally, residual stresses and slight variations in material properties may influence the measured response. The use of nominal values for Young's modulus and Poisson's ratio introduces further uncertainty.

The strain gauge method with a pressure range of 270-340 kPa while the diameter method with an average internal pressure of 305.6 ± 55.9 kPa. These values are consistent with typical internal pressures of carbonated beverage cans, generally ranging between 250-380 kPa [5]. Despite the large uncertainty for the diameter method, it falls within the range from the strain gauge method, suggesting that the method is valid in principle but limited by measurement precision. The agreement between experimental results and literature values indicated that the methods used provided a reasonable estimate of internal pressure.

Results from other lab groups were used to generate an average overall pressure as well as average pressure from hoop and longitudinal strain. Data below 100 kPa was deemed to be an outlier as an unrealistic value given the parameters of the experiment. The overall mean pressure was 290.71 ± 39.70 kPa with a 95% confidence interval of (251.01, 330.1) kPa. The hoop mean pressure was 288.36 ± 59.71 kPa with a 95% confidence interval of (228.63, 348.04) kPa and longitudinal mean pressure was 294.57 ± 58.96 kPa with a 95% confidence interval of (235.61, 353.53) kPa. The hoop CI was higher than the longitudinal CI as there were more samples for it. Our hoop and longitudinal pressure values fall within their respective 95% confidence intervals, meaning our values agree with across lab data. The diameter method falls within the average pressure 95% confidence interval, meaning it agrees with the across lab data. The diameter method is compared to the overall average since the pressure is found from only one strain value.

Limitations and Improvements

Longitudinal strain is highly sensitive to electrical noise due to its low magnitude, which in turn reduces the reliability in longitudinal pressure calculation. Similarly, the diameter change method was limited by the small change in diameter being measured, which approaches the resolution limit of standard measuring tools such as calipers. Using the thin-walled cylindrical pressure vessel assumption introduces modeling error, as the actual can geometry includes end caps and thickness variation.

To reduce electrical noise, the implementation of signal filtering or the use of higher quality instrumentation would improve the reliability of the longitudinal pressure calculation. Using a micrometer instead of calipers would increase the precision of the diameter measurement and improve reliability of the diameter-based method. Furthermore, using a more accurate model which would include the can's domed top, shaped bottom, and transition regions near the ends would increase the accuracy of the results compared to the thin-walled cylindrical assumption. Applying shell theory or finite element analysis would provide a more realistic representation of the stress distribution and improve the overall accuracy of the pressure estimation.

V. Conclusion

The objective of this experiment was to determine the internal pressure of a carbonated beverage assuming thin-walled cylindrical pressure vessel via two methods, using strain gauge measurement and a diameter change method. The strain gauge method produced pressure estimates of 340 kPa from hoop stress and 270 kPa from longitudinal stress, with a combined pressure of 320 kPa. These results are consistent with one another, literature values, and class-wide values. The diameter change method resulted in an average internal pressure of 305.63 ± 55.94 kPa. This is consistent with the strain gauge method, literature values, and class-wide values. Overall, the agreement between both methods and expected pressure ranges demonstrates that internal pressure can reasonably be estimated using deformation-based approaches. While the strain gauge method provided more direct measurement of deformation, the diameter change method served as a valid comparative technique despite its high uncertainty.

Appendix

A. Direct and Given Uncertainties

To determine how measurement error spread into the computed strain, stress, and pressure values, an uncertainty analysis was carried out. The measurement tools, LabVIEW output, and designated material-property sources were the sources of direct measurement uncertainties. When collecting repeated measurements, statistical uncertainty was employed. Table XIII provides a summary of the direct and given uncertainty used in the propagation.

TABLE XIII
DIRECT AND GIVEN UNCERTAINTIES

Quantity	Symbol	Uncertainty	Unit	Source
Thickness	μt	± 0.001	mm	Manufacturer (Micrometer)

TABLE XIII CONTINUED
DIRECT AND GIVEN UNCERTAINTIES

Quantity	Symbol	Uncertainty	Unit	Source
Elastic Modulus	μE	± 6.9	GPa	[3]
Poisson's Ratio*	μv	-	-	[3]
Gauge Factor	μG_f	± 0.02	-	Given
Gain	μG	± 4.5	-	Given
Diameter Closed	μD_o	± 0.005	mm	Manufacturer (Caliper)
Diameter Opened	μD_f	± 0.005	mm	Manufacturer (Caliper)
Vs Longitudinal Open	μV_{sLo}	± 0.0002	V	95% CI
Vs Hoop Open	μV_{sHo}	± 0.0002	V	95% CI
Vs Longitudinal Closed	μV_{sLc}	± 0.0002	V	95% CI
Vs Hoop Closed	μV_{sHc}	± 0.0002	V	95% CI
Vamp Longitudinal Open	μV_{ampLo}	± 0.00750	V	95% CI
Vamp Hoop Open	μV_{ampHo}	± 0.00771	V	95% CI
Vamp Longitudinal Closed	μV_{ampLc}	± 0.00754	V	95% CI
Vamp Hoop Closed	μV_{ampHc}	± 0.00765	V	95% CI
Stress Longitudinal	$\mu\sigma_L$	± 9.32	MPa	RSS
Stress Hoop	$\mu\sigma_H$	± 18.34	MPa	RSS
Strain Hoop	$\mu\varepsilon_H$	± 0.00124	mm/mm	RSS
Strain Longitudinal	$\mu\varepsilon_L$	± 0.00005	mm/mm	RSS
Strain Diameter	$\mu\varepsilon_D$	± 0.01086	mm/mm	RSS

*Poisson's Ratio was reported as 0.34 and treated as constant for RSS propagations.

B. *Strain Gauge Method RSS Propagation*

The strain-gauge method first converted amplified bridge voltage into strain. For each strain gauge, strain was calculated using the amplified quarter-bridge relation equation (4). Because this strain value depends on several uncertain quantities, the uncertainty in strain was propagated using equation (12).

$$\mu_\varepsilon = \left[\left(\mu_{\Delta Vamp}\left(\frac{\partial\varepsilon}{\partial\Delta V_{amp}}\right)\right)^2 + \left(\mu_{Vs}\left(\frac{\partial\varepsilon}{\partial V_s}\right)\right)^2 + \left(\mu_{Gf}\left(\frac{\partial\varepsilon}{\partial G_f}\right)\right)^2 + \left(\mu_{Gain}\left(\frac{\partial\varepsilon}{\partial Gain}\right)\right)^2\right]^{\frac{1}{2}} \tag{12}$$

Where μ is the uncertainty of its corresponding variable. This propagation was applied to the hoop and longitudinal gauge readings for both the unopened and opened can states.

The pressure calculation used the change in strain between the sealed and opened can states. The strain difference was calculated as equation (5).

$$\mu_{\Delta\varepsilon} = \left[\left(\mu_{\varepsilon_f}\frac{\partial\Delta\varepsilon}{\partial\varepsilon_f}\right)^2 + \left(\mu_{\varepsilon_o}\frac{\partial\Delta\varepsilon}{\partial\varepsilon_o}\right)^2\right]^{\frac{1}{2}} \tag{13}$$

The hoop and longitudinal strain changes were then converted into stresses using the plane-stress relations seen in equations (6)-(7). Where σ_H is hoop stress, σ_L is longitudinal stress, E is Young's modulus, ε_H is hoop strain, and ε_L is longitudinal strain. The uncertainty in each stress value was propagated using the general root-sum-square in (14)-(15).

$$\mu_{\sigma_H} = \left[\left(\mu_{\varepsilon_H}\frac{\partial\sigma_H}{\partial\varepsilon_H}\right)^2 + \left(\mu_{\varepsilon_L}\frac{\partial\sigma_H}{\partial\varepsilon_L}\right)^2 + \left(\mu_E\frac{\partial\sigma_H}{\partial E}\right)^2\right]^{\frac{1}{2}} \tag{14}$$

$$\mu_{\sigma_L} = \left[\left(\mu_{\varepsilon_L}\frac{\partial\sigma_L}{\partial\varepsilon_L}\right)^2 + \left(\mu_{\varepsilon_H}\frac{\partial\sigma_L}{\partial\varepsilon_H}\right)^2 + \left(\mu_E\frac{\partial\sigma_L}{\partial E}\right)^2\right]^{\frac{1}{2}} \tag{15}$$

The combined hoop and longitudinal stress state was used to calculate the final strain-gauge pressure estimate (3). The pressure equation was expressed as follows because the report utilized can diameter rather than radius.

$$\mu_P = \left[\left(\mu_{\sigma_H}\frac{\partial P}{\partial\sigma_H}\right)^2 + \left(\mu_{\sigma_L}\frac{\partial P}{\partial\sigma_L}\right)^2 + \left(\mu_t\frac{\partial P}{\partial t}\right)^2 + \left(\mu_d\frac{\partial P}{\partial d}\right)^2\right]^{\frac{1}{2}} \tag{16}$$

The final strain gauge pressure estimate presented in the Results and Discussion is supported by this propagation. Voltage noise, strain gauge sensitivity, and the lesser magnitude of the longitudinal strain signal were the key factors influencing the uncertainty of the strain-gauge approach since it directly measured the local mechanical reaction of the can wall.

C. *Change of Diameter Propagation*

The change in can diameter before and after opening was used to calculate the second pressure estimate. The can wall contracted somewhat and the interior pressure fell to air pressure when the can was opened. Circumferential strain was estimated using this diameter change in equation (8).

$$\mu_{\varepsilon_D} = \left[\left(\mu_{D_o} \frac{\partial \varepsilon_D}{\partial D_o} \right)^2 + \left(\mu_{D_f} \frac{\partial \varepsilon_D}{\partial D_f} \right)^2 \right]^{\frac{1}{2}} \tag{17}$$

Since the diameter strain represents circumferential strain, it was combined with the thin-walled pressure-vessel relations to estimate pressure. The diameter-method pressure propagation was calculated by (18) using equation (9).

$$\mu_{P_D} = \left[\left(\mu_{\varepsilon_D} \frac{\partial P_D}{\partial \varepsilon_D} \right)^2 + \left(\mu_{D_o} \frac{\partial P_D}{\partial D_o} \right)^2 + \left(\mu_t \frac{\partial P_D}{\partial t} \right)^2 + \left(\mu_E \frac{\partial P_D}{\partial E} \right)^2 \right]^{\frac{1}{2}} \tag{18}$$

Due to the close values of Do and Df, the diameter-change approach is susceptible to measurement error. As a result, the pressure estimate is based on a little discrepancy between two comparable observations. ε_D and P_D can be significantly impacted by small variations in caliper alignment, applied caliper force, or local can deformation. When comparing the diameter-method results to the strain-gauge approach, this restriction is crucial.

D. *Final Propagated Uncertainties*

Table XIV provides a summary of the final propagated pressure estimates from both approaches. The purpose of this study was to ascertain if the diameter-change and strain-gauge approaches yielded reliable estimates of the initial can pressure and to pinpoint the primary source of error for each approach.

TABLE XIV
PRESSURE UNCERTAINTIES

Quantity	Symbol	Uncertainty	Unit	Source
Pressure Longitudinal	μ_{PL}	± 78.61	kPa	RSS
Pressure Hoop	μ_{PH}	± 104.68	kPa	RSS
Pressure	μ_P	± 11.27	kPa	RSS
Pressure Class	μ_{Pc}	± 78.61	kPa	95% CI
Pressure Class Longitudinal	μ_{PcL}	± 58.96	kPa	95% CI
Pressure Class Hoop	μ_{PcH}	± 59.705	kPa	95% CI
Pressure Diameter	μ_{PD}	± 55.94	kPa	RSS

Overall, the uncertainty propagation demonstrates that, although having distinct measurement constraints, both approaches used the same pressure release to estimate the can pressure. The diameter-change approach calculated pressure from the global geometric change following opening, whereas the strain-gauge method assessed pressure-induced deformation directly through can wall strain. The final pressure estimate is supported by agreement between the two approaches, but neither approach should be regarded as precise due to the various sources of error.

E. *Total Class Data*

In order to compare the group's results with the wider class dataset, pressure measurements from the other lab sections were added after the Group 31 pressure values were computed. Because each group tested a different can, employed a different gauge installation, and carried out its own measuring process, these values were regarded as independent pressure estimations rather than repeated trials. The class data was not integrated with the Group 31 values for the primary pressure calculation; rather, they were utilized as a comparison dataset using equations (19)-(24).

$$\bar{x} = \frac{1}{N} \cdot \sum_{i=1}^{N} x_i \tag{19}$$

$$S_x = \sqrt{\frac{1}{N-1} \cdot \sum_{i=1}^{N} (x_i - \bar{x})^2} \tag{20}$$

$$S_{\bar{x}} = \frac{S_x}{\sqrt{N}} \tag{21}$$

$$P_x = tS_x \tag{22}$$

$$CI = \bar{x} \pm P_{\bar{x}} \tag{23}$$

$$Bounds = \bar{x} \pm 2S_x \tag{24}$$

Using the final retained class dataset, the mean, standard deviation, standard error, confidence interval, and bounds were evaluated for each pressure-estimation method. Any excluded group result was kept visible in the class data table and marked with the reason for exclusion in Table XV. The purpose of this subsection is to document the statistical spread of the total class pressure results and define the class-level comparison values referenced in the Discussion.

TABLE XV
TOTAL CLASS PRESSURE DATA USED FOR COMPARISON

Pressure [Pa]	Orientation	Soda
1974.050648*	Axial	Dr. Pepper
6700*	Axial	Coke
11515.34*	Axial	Coke
234052	Axial	Dr. Pepper
238673	Axial	Coke
244946	Axial	Dr. Pepper
270017.0503	Axial	Dr. Pepper
274105.3	Axial	Dr. Pepper
284787	Axial	Coke
399000	Axial	Coke
410949.1354	Axial	Coke
147000	Hoop	Coke
158000	Hoop	Coke

TABLE XV CONTINUED
TOTAL CLASS PRESSURE DATA USED FOR COMPARISON

Pressure [Pa]	Orientation	Soda
189963.7244	Hoop	Dr. Pepper
222424	Hoop	Coke
229750.6273	Hoop	Dr. Pepper
253372.07	Hoop	Coke
300918.24	Hoop	Dr. Pepper
310860	Hoop	Coke
338000	Hoop	Coke
349325.732	Hoop	Coke
374711.713	Hoop	Coke
398435	Hoop	Dr. Pepper
475600	Hoop	Dr. Pepper

*Indicates outliers that were removed from calculations.

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